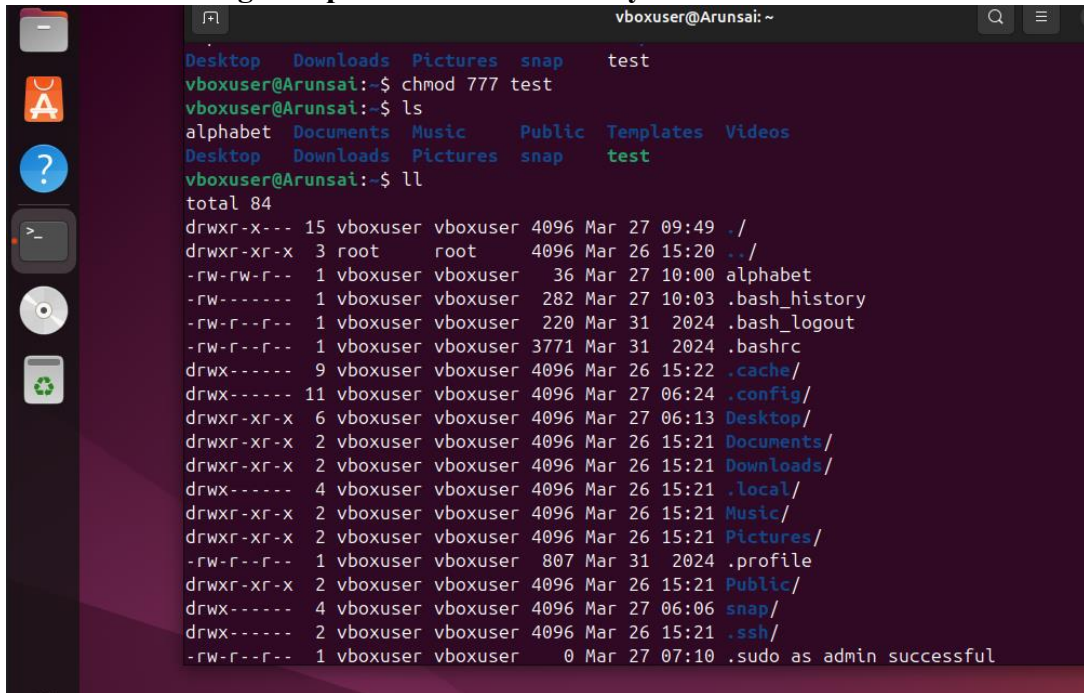


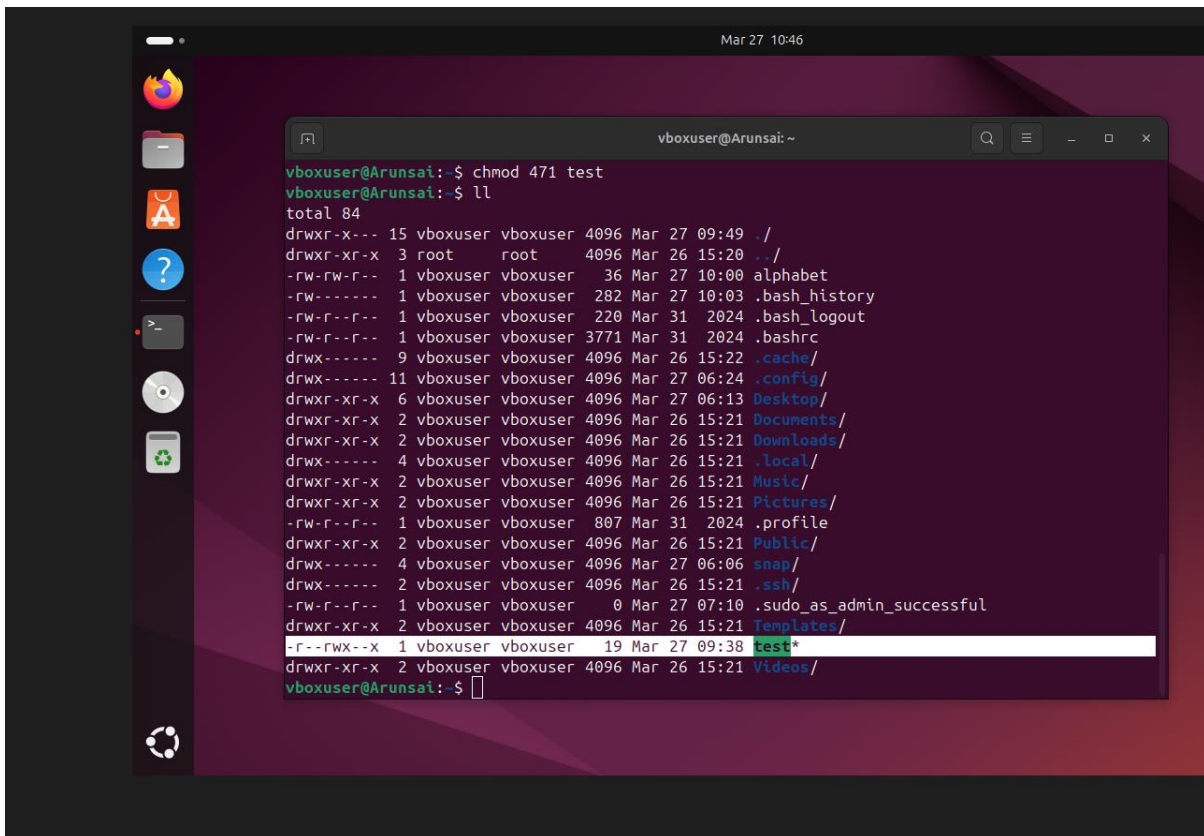
Linux 02 Commands

- **Chmod: To change the permission of directory file**



```
vboxuser@Arunasai: ~  
Desktop Downloads Pictures snap test  
vboxuser@Arunasai:~$ chmod 777 test  
vboxuser@Arunasai:~$ ls  
alphabet Documents Music Public Templates Videos  
Desktop Downloads Pictures snap test  
vboxuser@Arunasai:~$ ll  
total 84  
drwxr-x--- 15 vboxuser vboxuser 4096 Mar 27 09:49 ./  
drwxr-xr-x  3 root      root      4096 Mar 26 15:20 ../  
-rw-rw-r--  1 vboxuser vboxuser   36 Mar 27 10:00 alphabet  
-rw-----  1 vboxuser vboxuser  282 Mar 27 10:03 .bash_history  
-rw-r--r--  1 vboxuser vboxuser  220 Mar 31 2024 .bash_logout  
-rw-r--r--  1 vboxuser vboxuser 3771 Mar 31 2024 .bashrc  
drwx-----  9 vboxuser vboxuser 4096 Mar 26 15:22 .cache/  
drwx----- 11 vboxuser vboxuser 4096 Mar 27 06:24 .config/  
drwxr-xr-x  6 vboxuser vboxuser 4096 Mar 27 06:13 Desktop/  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Documents/  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Downloads/  
drwx-----  4 vboxuser vboxuser 4096 Mar 26 15:21 .local/  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Music/  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Pictures/  
-rw-r--r--  1 vboxuser vboxuser  807 Mar 31 2024 .profile  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Public/  
drwx-----  4 vboxuser vboxuser 4096 Mar 27 06:06 snap/  
drwx-----  2 vboxuser vboxuser 4096 Mar 26 15:21 .ssh/  
-rw-r--r--  1 vboxuser vboxuser   0 Mar 27 07:10 .sudo_as_admin_successful
```

- **Command : Chmod 471 test**
- **Read, Write, Execute = 4, 2, 1 (Numeric Numbers)**



```
Mar 27 10:46  
vboxuser@Arunasai: ~  
vboxuser@Arunasai:~$ chmod 471 test  
vboxuser@Arunasai:~$ ll  
total 84  
drwxr-x--- 15 vboxuser vboxuser 4096 Mar 27 09:49 ./  
drwxr-xr-x  3 root      root      4096 Mar 26 15:20 ../  
-rw-rw-r--  1 vboxuser vboxuser   36 Mar 27 10:00 alphabet  
-rw-----  1 vboxuser vboxuser  282 Mar 27 10:03 .bash_history  
-rw-r--r--  1 vboxuser vboxuser  220 Mar 31 2024 .bash_logout  
-rw-r--r--  1 vboxuser vboxuser 3771 Mar 31 2024 .bashrc  
drwx-----  9 vboxuser vboxuser 4096 Mar 26 15:22 .cache/  
drwx----- 11 vboxuser vboxuser 4096 Mar 27 06:24 .config/  
drwxr-xr-x  6 vboxuser vboxuser 4096 Mar 27 06:13 Desktop/  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Documents/  
drwx-----  4 vboxuser vboxuser 4096 Mar 26 15:21 .local/  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Music/  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Pictures/  
-rw-r--r--  1 vboxuser vboxuser  807 Mar 31 2024 .profile  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Public/  
drwx-----  4 vboxuser vboxuser 4096 Mar 27 06:06 snap/  
drwx-----  2 vboxuser vboxuser 4096 Mar 26 15:21 .ssh/  
-rw-r--r--  1 vboxuser vboxuser   0 Mar 27 07:10 .sudo_as_admin_successful  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Templates/  
-r--rwx--x  1 vboxuser vboxuser   19 Mar 27 09:38 test*  
drwxr-xr-x  2 vboxuser vboxuser 4096 Mar 26 15:21 Videos/  
vboxuser@Arunasai:~$
```

- Chown root:root filename to change owner name

```
vboxuser@Arunsai:~/Desktop$ sudo chown root:root Arun
/etc/sudoers:2:2: syntax error

total 28
drwxr-xr-x  6 vboxuser vboxuser 4096 Mar 27 06:13 ./
drwxr-x--- 15 vboxuser vboxuser 4096 Mar 27 09:49 ../
drwxrwxrwx  2 root      root    4096 Mar 26 15:23 Arun/
drwxrwxr-x  2 vboxuser vboxuser 4096 Mar 26 15:25 Chaitanya/
drwxrwxr-x  3 vboxuser vboxuser 4096 Mar 27 06:13 d1/
```

- And changed again to previous owner

```
drwxrwxr-x  2 vboxuser vboxuser 4096 Mar 26 15:24 Ganesh/
vboxuser@Arunsai:~/Desktop$ sudo chown vboxuser:vboxuser Arun

vboxuser@Arunsai:~/Desktop$ ll
total 28
drwxr-xr-x  6 vboxuser vboxuser 4096 Mar 27 06:13 ./
drwxr-x--- 15 vboxuser vboxuser 4096 Mar 27 09:49 ../
drwxrwxrwx  2 vboxuser vboxuser 4096 Mar 26 15:23 Arun/
drwxrwxr-x  2 vboxuser vboxuser 4096 Mar 26 15:25 Chaitanya/
drwxrwxr-x  3 vboxuser vboxuser 4096 Mar 27 06:13 d1/
-rw-rw-r--  1 vboxuser vboxuser   41 Mar 26 15:31 file
drwxrwxr-x  2 vboxuser vboxuser 4096 Mar 26 15:24 Ganesh/
```

- cat /etc/passwd : list of users in VM or Server

```
vboxuser@Arunsai:~/Desktop$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
bin:x:39:39:bin:/bin:/usr/sbin/nologin
```

- Switch user sudo su -
- Id: It is display what we used

```
vboxuser@Arunsai:~$ uname
Linux
vboxuser@Arunsai:~$ id
uid=1000(vboxuser) gid=1000(vboxuser) groups=1000(vboxuser),27(sudo)
vboxuser@Arunsai:~$ su root
Password:
su: Authentication failure
vboxuser@Arunsai:~$ sudo su -
root@Arunsai:~# id
uid=0(root) gid=0(root) groups=0(root)
root@Arunsai:~#
```

- Useradd filename:
- Passwd:
- New password:

```
vboxuser:x:1000:1000:vboxuser:./home/vboxuser:./bin/bash
Arun:x:1001:1001:./home/Arun:/bin/sh
root@Arunsai:~# useradd Arun^c
root@Arunsai:~# passwd Arun
New password:
```

- Successfully new password created

```
passwd: password updated successfully
root@Arunsai:~#
```

- Find / -name sudoers: to find which are related to sudoers

```
Arun is not in the sudoers file.
$ exit
root@Arunsai:~# find / -name sudoers
/etc/sudoers
/usr/share/doc/sudo/examples/sudoers
```

- Cat /etc/sudoers: to give permissions

```

/etc/sudoers
/usr/share/doc/sudo/examples/sudoers
/snap/core22/2292/etc/sudoers
find: '/run/user/1000/gvfs': Permission denied
find: '/run/user/1000/doc': Permission denied
root@Arun sai:~# cat /etc/sudoers
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults      env_reset
Defaults      mail_badpass
Defaults      secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/
snap/bin"

```

- **vi /etc/suoders:**

```

root@Arun sai: ~
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults      env_reset
Defaults      mail_badpass
Defaults      secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/
snap/bin"
# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults      use_pty
# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"
# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
W10: Warning: Changing a readonly file

```

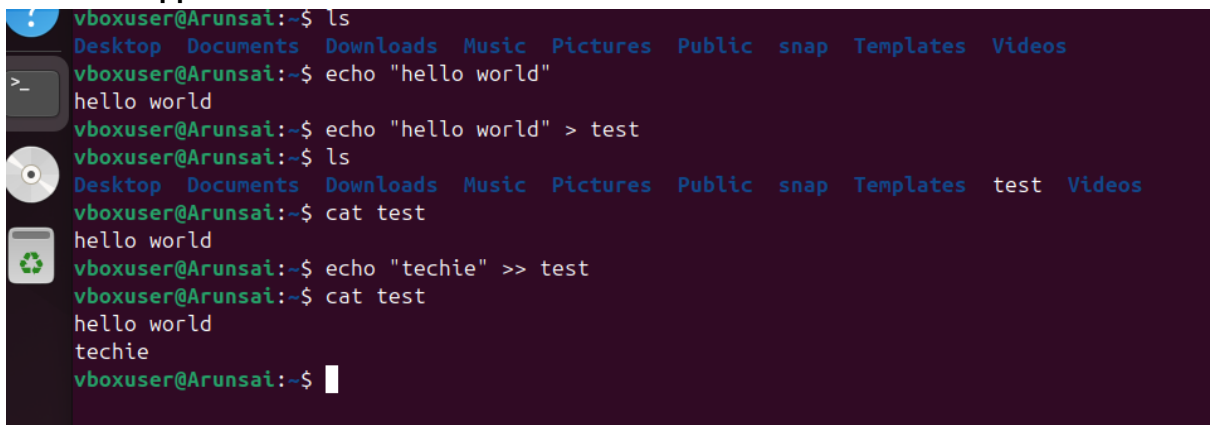
- **Given permissions to suoders**

```

# Allow members of group sudo to execute any command
%sudo    ALL=(ALL:ALL) ALL
Arun ALL=(ALL:ALL) ALL

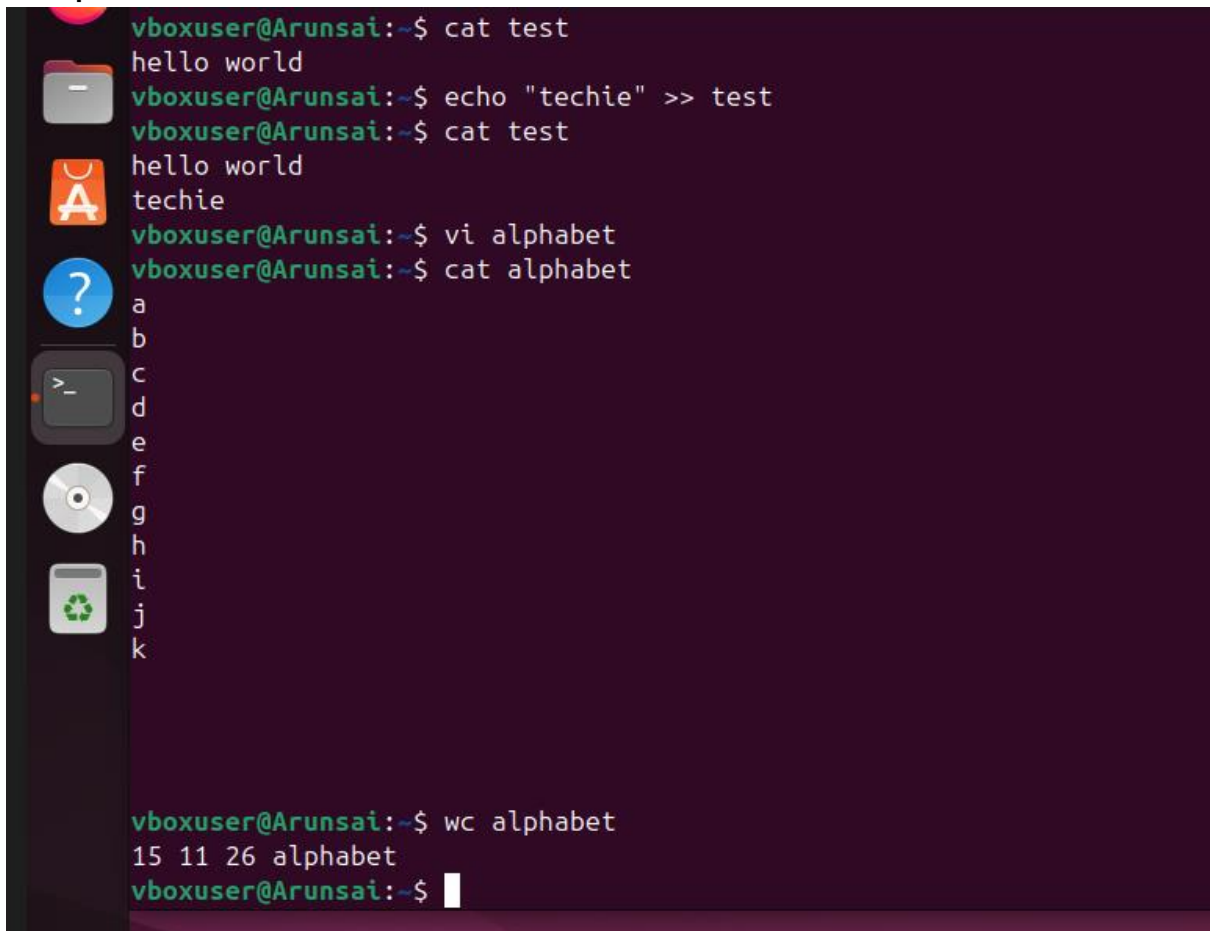
```

- Redirection of output: echo "hello world" > test
- Double or append redirection : echo "techie" >> test



```
vboxuser@Arunsai:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
vboxuser@Arunsai:~$ echo "hello world"
hello world
vboxuser@Arunsai:~$ echo "hello world" > test
vboxuser@Arunsai:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  test  Videos
vboxuser@Arunsai:~$ cat test
hello world
vboxuser@Arunsai:~$ echo "techie" >> test
vboxuser@Arunsai:~$ cat test
hello world
techie
vboxuser@Arunsai:~$
```

- vi alphabet : to edit
- cat alphabet: to read
- wc alphabet: to check the word count



```
vboxuser@Arunsai:~$ cat test
hello world
vboxuser@Arunsai:~$ echo "techie" >> test
vboxuser@Arunsai:~$ cat test
hello world
techie
vboxuser@Arunsai:~$ vi alphabet
vboxuser@Arunsai:~$ cat alphabet
a
b
c
d
e
f
g
h
i
j
k

vboxuser@Arunsai:~$ wc alphabet
15 11 26 alphabet
vboxuser@Arunsai:~$
```

- head -3 alphabet: call first 3 alphabets
- tail -8 alphabet: call the last 8 alphabets

```
vboxuser@Arunsa:~$ wc alphabet
15 11 26 alphabet
vboxuser@Arunsa:~$ head -3 alphabet
a
b
c
vboxuser@Arunsa:~$
```

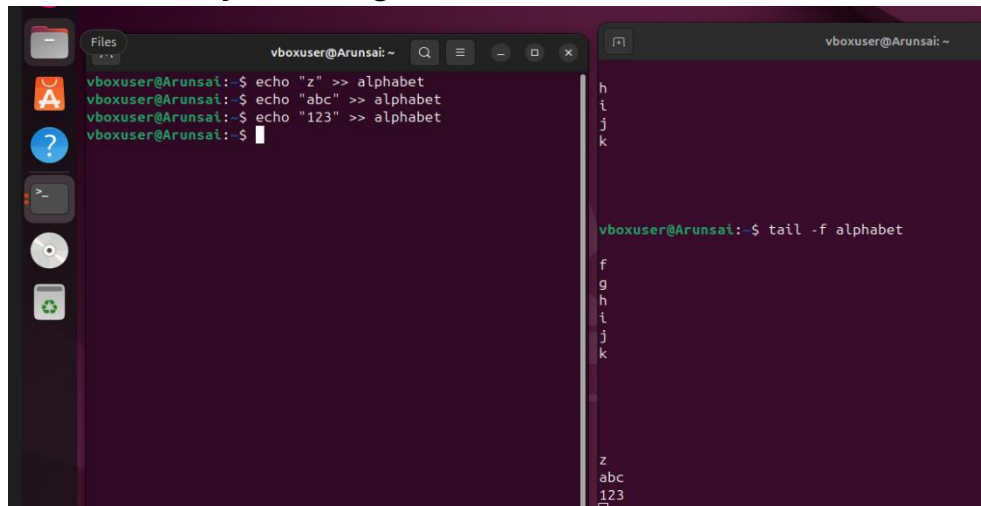
```
vboxuser@Arunsa:~$ tail -8 alphabet
h
i
j
k

vboxuser@Arunsa:~$
```

- **tail -f alphabet** : call from the f character to end

```
vboxuser@Arunsa:~$ tail -f alphabet
f
g
h
i
j
k
```


- It is continuously monitoring the file



The screenshot shows a terminal window with a dark purple background. On the left, there is a sidebar with icons for Files, Applications, and a search icon. The terminal displays the following commands and output:

```
vboxuser@Arunsai:~$ echo "z" >> alphabet
vboxuser@Arunsai:~$ echo "abc" >> alphabet
vboxuser@Arunsai:~$ echo "123" >> alphabet
vboxuser@Arunsai:~$
```

On the right side of the terminal, the output of the `tail -f alphabet` command is shown, displaying the content of the file as it is updated:

```
h
i
j
k

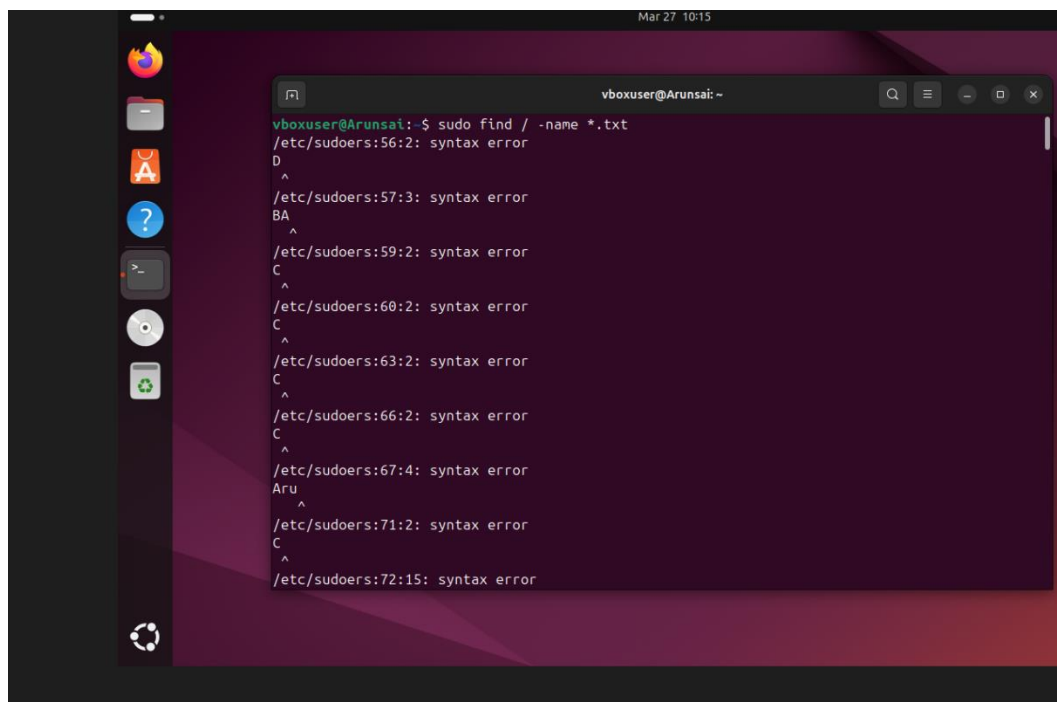
vboxuser@Arunsai:~$ tail -f alphabet

f
g
h
i
j
k

z
abc
123

```

- Sudo find / -name *.txt : to check total files



The screenshot shows a terminal window with a dark purple background. The terminal displays the output of the command `sudo find / -name *.txt`:

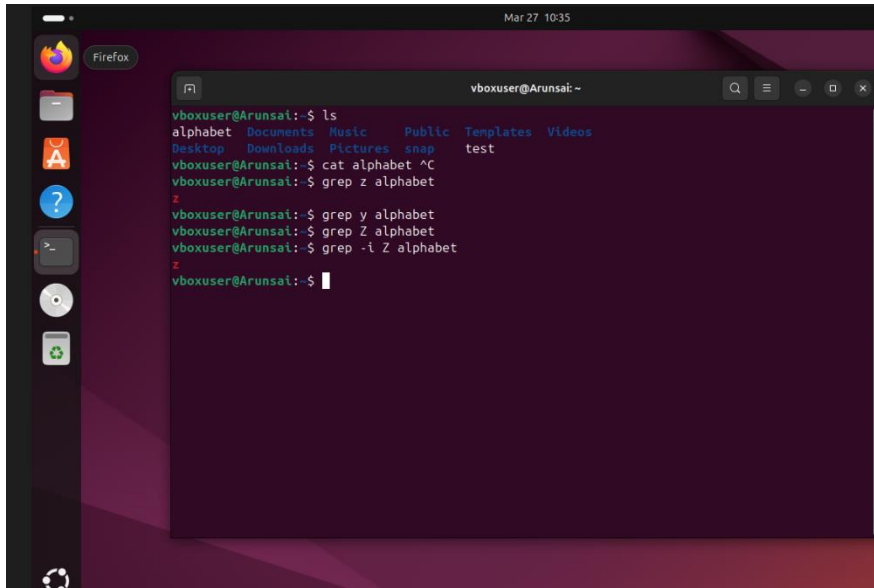
```
vboxuser@Arunsai:~$ sudo find / -name *.txt
/etc/sudoers:56:2: syntax error
D
^
/etc/sudoers:57:3: syntax error
BA
^
/etc/sudoers:59:2: syntax error
C
^
/etc/sudoers:60:2: syntax error
C
^
/etc/sudoers:63:2: syntax error
C
^
/etc/sudoers:66:2: syntax error
C
^
/etc/sudoers:67:4: syntax error
Aru
^
/etc/sudoers:71:2: syntax error
C
^
/etc/sudoers:72:15: syntax error
```

- Find / -name sudoers -mtime +30 : last 30 days files check

```
vboxuser@Arunsa: ~
vboxuser@Arunsa:~$ find / -name sudoers -mtime +30
find: '/etc/credstore': Permission denied
find: '/etc/ssl/private': Permission denied
find: '/etc/cups/ssl': Permission denied
find: '/etc/credstore.encrypted': Permission denied
find: '/etc/chatscripts': Permission denied
find: '/etc/ppp/peers': Permission denied
find: '/etc/ssh': Permission denied
find: '/etc/tmpn7exw8sg': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/lost+found': Permission denied
find: '/tmp/systemd-private-faf80a3175d74a349cd52dbc204a9b63-systemd-logind.service-FZxJH4':
Permission denied
find: '/tmp/systemd-private-faf80a3175d74a349cd52dbc204a9b63-systemd-resolved.service-Y8ZXAw':
Permission denied
find: '/tmp/snap-private-tmp': Permission denied
find: '/tmp/systemd-private-faf80a3175d74a349cd52dbc204a9b63-upower.service-nZUoSR': Permissi
on denied
find: '/tmp/systemd-private-faf80a3175d74a349cd52dbc204a9b63-fwupd.service-YrSAXu': Permissio
n denied
find: '/tmp/systemd-private-faf80a3175d74a349cd52dbc204a9b63-systemd-timesyncd.service-YhgTp':
Permission denied
find: '/tmp/systemd-private-faf80a3175d74a349cd52dbc204a9b63-colord.service-o2m8rm': Permissi
on denied
```

```
vboxuser@Arunsa: ~
^
/etc/sudoers:63:2: syntax error
C
^
/etc/sudoers:66:2: syntax error
C
^
/etc/sudoers:67:4: syntax error
Aru
^
/etc/sudoers:71:2: syntax error
C
^
/etc/sudoers:72:15: syntax error
Arun ALL=(ALL:
^
/etc/sudoers:73:12: syntax error
Arun ALL=()
^
/etc/sudoers:75:9: syntax error
Arun ALL()
^
/snap/core22/2292/etc/sudoers
find: '/run/user/1000/gvfs': Permission denied
find: '/run/user/1000/doc': Permission denied
vboxuser@Arunsa:~$
```


- **Grep** : It is help us to check the specific keyword in entire file



The screenshot shows a terminal window titled 'vboxuser@Arunsai: ~' with a search icon and window controls. The terminal output is as follows:

```
vboxuser@Arunsai:~$ ls
alphabet Documents Music Public Templates Videos
Desktop Downloads Pictures snap test
vboxuser@Arunsai:~$ cat alphabet ^C
vboxuser@Arunsai:~$ grep z alphabet
z
vboxuser@Arunsai:~$ grep y alphabet
vboxuser@Arunsai:~$ grep Z alphabet
vboxuser@Arunsai:~$ grep -i Z alphabet
Z
vboxuser@Arunsai:~$
```

The terminal window is part of a desktop environment with a dark theme. A sidebar on the left contains icons for Firefox, a file manager, and other applications. The top of the window shows the date and time as 'Mar 27 10:35'.